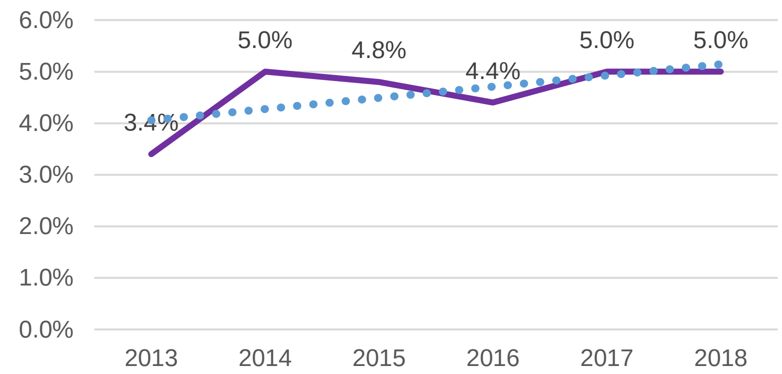
DEMAND ANALYSIS FOR ROOFTOP SOLAR SYSTEMS

A. Background

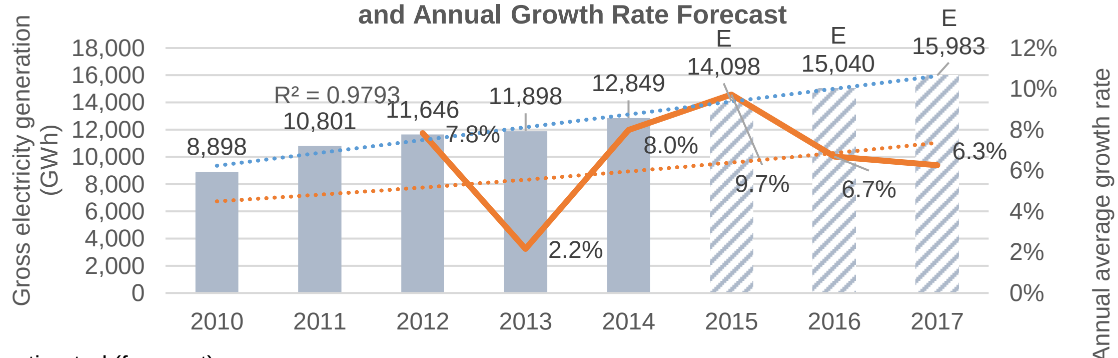
1. **Economic and Power Sector Growth.** Sri Lanka's economy has been growing steadily over the past few years, with gross domestic product averaging at 4.5% per annum from 2013-2017.[1] This growth prospect is expected to continue (Chart 1). The demand for energy generation is expected to grow at a slightly faster rate over the project implementation period from 2018–2022, as estimated from the trend analysis of gross electricity generation (Chart 2). Because electricity generation from fossil fuels accounts for 55% of the total energy generation mix,[2] resulting in high fuel import cost, Sri Lanka has been actively diversifying to renewable energy sources to ensure its energy security while improving the environmental conditions. Sri Lanka has substantial solar energy potential: for over two-thirds of the land mass, solar radiation varies from 4.0–4.5 kilowatt-hours per square meter per day (kWh/m^2/day), which is considered favourable for solar energy generation.[3]

Chart 1: Sri Lanka Annual GDP Growth Rate



Source: Asian Development Outlook 2017

Chart 2: Sri Lanka's Gross Electricity Generation and Annual Growth Rate Forecast



E = estimated (forecast)
Source: Sri Lanka Energy Balance 2014

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[1] ADB. 2017. *Asian Development Outlook.* Manila.

[2] Sri Lanka's energy generation mix is 55% thermal, 34% hydro, and 11% renewable, with solar power less than 1% of the total.

[3] Sri Lanka Sustainable Energy Authority. Solar Potential. http://www.energy.gov.lk/renewables/renewable-energy-resources/solar/solar-potential

2. Effective demand over the next few years is therefore likely to reflect (i) a combination of income effect reflecting the faster growth in energy demand commensurate with the growth in the economy; and (ii) a substitution effect reflecting a switchover of power generation to renewable energy from fossil fuel, further driven by technology-induced cost reduction in the renewable space.

3. **Strong Government Support.** The development of solar and rooftop solar power generation was based on the Government of Sri Lanka's (the government) strong policy initiative. In September 2016, the government announced "The Battle for Solar Energy" program,[4] under which the government intended to increase solar photovoltaic generation capacity from the current level of about 61.4 megawatts (MW) to 200.0 MW by 2020 and 1,000.0 MW by 2025. The Prime Minister's Office and the Ministry of Power and Renewable Energy led the rooftop solar energy development initiative. The strong government commitment to create a conducive business environment to enable rooftop PV system financing is the bedrock of this project design.

B. Market Condition

4. **Rooftop Solar Business Models.** The program envisages boosting clean power generation through net metering, net accounting, and micro solar power producer (net metering-plus) schemes to connect rooftop solar PV installations to the network:

(i) **Net metering.** Introduced in 2008, this model allows customers to offset the consumed electricity with the electricity they generate from their rooftop solar photovoltaic systems. At the end of monthly billing cycles, if the customers consume more than they have generated from their rooftop solar systems, they are billed at the electricity tariff rates based on their net consumption level. If the customers generate more electricity than they consume, the electricity surplus balance is carried over to the next month. The surplus balances can be kept for up to 10 years and distribution utilities will not pay customers for any surplus electricity they generate.

(ii) **Net accounting.** Introduced in 2016, this model allows customers to be paid in cash for any surplus they generate at the end of their monthly billing cycles at the following rates: (a) year 1 to year 7 at SLRs22.0/kWh, and (b) year 8 to year 20 at SLRs15.5/kWh. Other features are the same as for net metering.

(iii) **Net plus.** Unlike the other two models, this allows customers to separate electricity consumption and electricity generation. The customers pay for the electricity they consume based on the existing tariff structure. At the same time, the customers can also sell whatever electricity they generate from their rooftop solar photovoltaic systems at the following rates: (a) year 1 to year 7 at SLRs22.0/kWh; and (b) year 8 to year 20 at SLRs15.5/kWh. This scheme was introduced in 2016.

5. **Current Market Condition.** The current market condition in Sri Lanka is becoming favourable for rooftop solar system development. The government introduced net metering in July 2008. Ceylon Electricity Board (CEB) currently has attractive rates of SLRs22.0 per kWh for 7 years and SLRs15.5 per kWh for the next 12 years. There are more than 100 dedicated private sector renewable energy service companies to provide rooftop solar systems installation and

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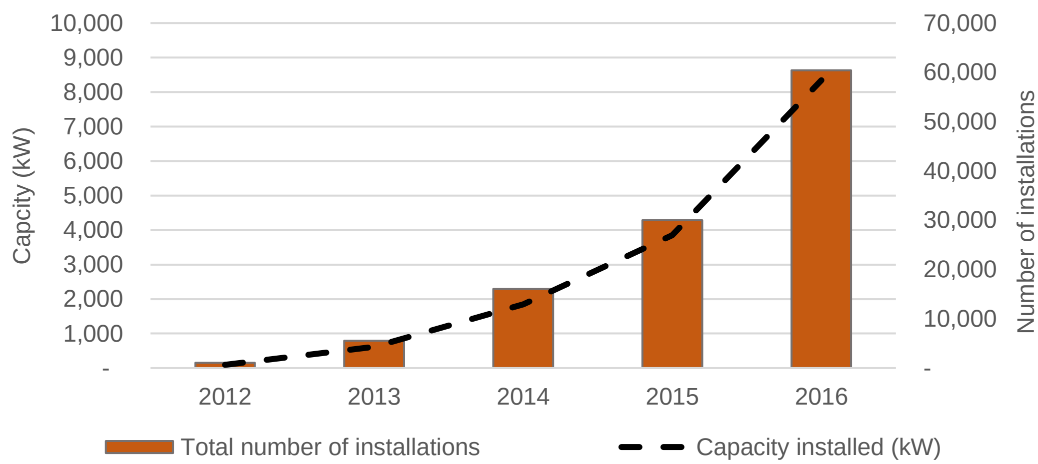
[4] Ministry of Power and Renewable Energy. Battle for Solar Energy begins from Presidents house. http://powermin.gov.lk/english/?p=4454

maintenance. Both solar panels and inverters enjoy duty and tax exemptions. Both the Government of Sri Lanka and Lanka Electricity Company (LECO) have subsidized rooftop solar loan schemes at 6% and 8% of annual interest rate (and a tenor about 5–7 years) to catalyse market demand. Both solar panels and inverters enjoy duty and tax exemptions. As a result, the rooftop solar power generation capacity expanded rapidly. The net metering model was introduced in 2008, and the net accounting and net plus models were only introduced in 2016. In 2014, the total installed capacity under the net metering scheme was about 14.4 MW. By the end of 2016, the installed rooftop solar capacity under the net metering scheme was 58 MW for 8,620 customers (e.g., household, institutions, and commercial entities); under the net accounting scheme is 2 MW for 694 customers, and under the net plus scheme is 1.4 MW for 88 entities. In 2017, an estimated 4,200 new rooftop solar photovoltaic system customers would be installed under the net metering scheme, adding to an additional generation capacity of about 30 MW.[5] About 95% of net metered customers are households, while the others are institutional customers. The recent expansion of the rooftop solar capacity is due to promotion policy particularly reduced lending rates based on the current system unit costs, which are low in historical terms. Net accounting and net plus schemes were both introduced in 2016 and there are limited track records to date.

Chart 3: Current Market Condition under Net Metering Scheme



Source: Asian Development Bank staff estimates

C. Market Analysis

6. **Basic Factors.** There are three key variables to affect the market demand: (i) existing electricity tariff rates; (ii) rooftop solar system costs; and (iii) financing incentives. First, it is assumed that existing electricity tariff rates are fixed and do not affect the customer incentive to install rooftop solar systems. Second, the system costs are also fixed on a per-unit basis (per kilowatt peak) over the course of ADB project implementation (e.g., $1,500 for residential

[5] The current estimate of the rooftop solar PV cumulative generating capacity is about 80 MW.

customers and $1,000 for industrial and commercial customers).[6] Third, the only meaningful variable to affect the rooftop solar system financing is the bank lending rates.

Table 1: Sri Lanka's Selected Electricity Tariff Rates[7]

Type of Customers	Monthly Consumption (kWh)	Energy Charge (SLRs/kWh)
Residential	0–30	2.50
	31–60	4.85
	61–90	10.00
	91–120	27.75
	121–180	32.00
	180 and above	45.00
Religious and Charitable	0–30	1.90
	31–90	2.80
	91–120	6.75
	121–180	7.50
	180 and above	9.40
Industrial	0–300	10.80
	300 and above	12.20
Hotel		21.50
General	0–300	18.30
	300 and above	22.85
Government		14.35–14.65

kWh = kilowatt-hour, SLR = Sri Lanka rupee.
Note: Excluding the fixed charge and maximum demand charge for clear illustration.
Source: Sri Lanka Public Utilities Commission.

(i) **Why net metering?** Residential customers with high electricity consumption, such as those who consume over 120 kWh with electricity tariff rates of SLRs32.0 per kWh and those who consume over 180 kWh with electricity tariff rates of SLRs45.0 per kWh (Table 1), should choose the net metering scheme. These customers can use the electricity generated from their rooftop solar systems with a cost base of SLRs20.00–SLRs30.00 per kWh (footnote 6) over the lifecycle of the system of 20 years to offset their high electricity tariff rates of SLRs32.00 and SLRs45.00 per kWh. The financial return to the subborrowers under the net metering model is 14.5%.

(ii) **Why net accounting?** Residential customers who consume between 91–120 kWh of electricity per month have a tariff rate of SLRs27.75 per kWh. Some non-residential customers (e.g., hotels and general business category) have similar electricity tariff rates between SLRs21.5 and SLRs22.85 per kWh (Table 1) should choose the net accounting scheme. At a lower electricity consumption level, these customers may produce some surplus electricity and

[6] Therefore, the total cost of any rooftop solar system is not a relevant factor for market demand. In addition, the average total cost of installing a household rooftop solar system (the CAPEX costs) is between $2,250 to $7,500, assuming $1,500 per kilowatt peak (one-off cost including solar panels, invertors, frames, and labour costs) and an average household rooftop solar system capacity between 1.5 kWp to 5.0 kWp. This translates to a total system unit cost between $0.13–$0.20 (SLRs20.00–SLRs30.00) per kWh during the entire lifecycle of the system.

[7] Available: http://www.pucsl.gov.lk/english/industries/electricity/electricity-tariffscharges

could sell it to the CEB and LECO at the rates of (a) SLRs22.0 per kWh at year 1 to year 7; and (b) at SLRs15.5 per kWh on year 8 to year 20. The financial return to the subborrowers under the net accounting model is 12.6%.

(iii) **Why net plus?** Residential customers who consume below 90 kWh of electricity per month with a tariff of below SLRs10.0 per kWh, and/or for non-residential customers with the lowest tariff brackets, should choose the net plus scheme. These customers, especially those with large rooftop spaces and economy of scales could benefit from selling all electricity generated from rooftop solar systems to the utilities at the rates of (a) SLRs22.0 per kWh at year 1 to year 7; and (b) SLRs15.5 per kWh year 8 to year 20. The financial return to the subborrowers under the net plus model is 21.8%.[8]

7. **Market Size for net metering.** Current market study estimates about 70,000 customers who would consume over 210 kWh per month in 2017 but have not installed any rooftop solar systems. Because they pay the highest electricity grid tariff rate at SLRs45 (see Table 1), this segment of customers would find it financially beneficial to install rooftop solar systems at existing bank lending rate of about 15% per annum with 5–7 years of tenor. Customers who consume less than 210 kWh would not be interested.[9] Assuming conservatively that 10% of those customers who consume over 210 kWh would be willing to install rooftop solar systems, and each system costs $5,000 (e.g., assuming conservatively of installing a small 3–3.5 kWp system), the potential customer base would be 7,000 and market demand for funds would be $35 million (Table 2).

Table 2: Potential Market Size under the Net Metering Scheme based on Existing 15% Bank Lending Rates (Residential)

Monthly Consumption (kWh)	**Existing Tariff** (SLR/unit)	**Total Existing Customers**	**Potential Solar PV Customers[a]**	**CAPEX Unit Cost**	**System Size** (kWp)	**System Cost per Customer**	**Market Size** ($ million)
210 and above	45	70,000	7,000 (10%)	$1,500	3.0–3.5	$5,000	35
181 – 210	45	200,000	-	-	-	-	-
121 – 180	30	500,000	-	-	-	-	-
91 – 120	28	1,000,000	-	-	-	-	-
31 – 90	10	1,500,000	-	-	-	-	-
0 – 30	5	1,800,000	-	-	-	-	-
Total	3	5,000,000	7,000	-	-	-	35

CAPEX = capital expenditure, kWh = kilowatt hour, kWp = kilowatt peak, PV = photovoltaic, SLR = Sri Lanka rupee.
Note: It is assumed that the total households (billed accounts) are 5,000,000, out of a population of 21,000,000 in Sri Lanka.
[a] The number of potential customers would be 7,000 (e.g., 15,000 total customers – 8,000 existing customers).

8. When the commercial bank lending rate for rooftop solar systems declines to 12%, more customers would be interested. Assuming conservatively that 15% of the customers who consume over 210 kWh of electricity per month and 5% of the customers who consumes between 181 to 210 kWh of electricity per month would find it financially beneficial and be willing to install rooftop solar systems, the total market demand for funds would be $70 million (see Table 3).

[8] However, the actual FIRR under net plus scheme will depend on the economy of scale.
[9] Although some customers who consume between 181 kWh to 210 kWh also pay SLRs45 per unit, their consumption of fewer units of electricity may not justify their savings from installing the rooftop solar systems.

Table 3: Potential Market Size under the Net Metering Scheme based on 12% Bank Lending Rates (Residential)

Monthly Consumption (kWh)	**Existing Tariff** (SLR/unit)	**Total Existing Customers**	**Potential Solar PV Customers**[a]	**CAPEX Unit Cost**	**System Size** (kWp)	**System Cost per Customer**	**Market Size** ($ million)
210 and above	45	70,000	10,500 (15%)	$1,500	3.0–3.5	$5,000	53
181 – 210	45	70,000	3,500 (5%)	$1,500	3.0–3.5	$5,000	17
121 – 180	30	400,000	-	-	-	-	-
91 – 120	28	800,000	-	-	-	-	-
31 – 90	10	1,660,000	-	-	-	-	-
0 – 30	5	2,000,000	-	-	-	-	-
Total	3	5,000,000	14,000	-	-	-	70

CAPEX = capital expenditure, kWh = kilowatt hour, kWp = kilowatt peak, PV = photovoltaic, SLR = Sri Lankan rupee.
Note: It is assumed that the total households (billed accounts) are 5,000,000, out of a population of 21,000,000 in Sri Lanka.
[a] The number of potential customers would be 7,000 (e.g., 15,000 total customers – 8,000 existing customers).

9. When the commercial bank lending rate declines to 10%, customers' demand for rooftop solar system financing would increase even further. Assuming conservatively that 20% of the customers who consumes over 210 kWh of electricity per month, 10% of the customers who consumes over between 181 kWh to 210 kWh of electricity per month, and 2% of the customers who consumes over between 121 kWh to 180 kWh of electricity per month would find it financially beneficial and be willing to install rooftop solar systems, the total market demand for funds would be $140 million.

Table 4: Potential Market Size under the Net Metering Scheme based on 10% Bank Lending Rates (Residential)

Monthly Consumption (kWh)	**Existing Tariff** (SLR/unit)	**Total Existing Customers**	**Potential Solar PV Customers**	**CAPEX Unit Cost**	**System Size** (kWp)	**System Cost per Customer**	**Market Size** ($ million)
210 and above	45	70,000	14,000 (20%)	$1,500	3.0–3.5	$5,000	70
181 – 210	45	70,000	7,000 (10%)	$1,500	3.0–3.5	$5,000	35
121 – 180	30	400,000	9,000 (2%)	$1,500	3.0–3.5	$5,000	45
91 – 120	28	800,000	-	-	-	-	-
31 – 90	10	1,660,000	-	-	-	-	-
0 – 30	5	2,000,000	-	-	-	-	-
Total	3	5,000,000	30,000	-	-	-	150

CAPEX = capital expenditure, kWh = kilowatt hour, kWp = kilowatt peak, PV = photovoltaic, SLR = Sri Lankan rupee.
Note: It is assumed that the total households (billed accounts) are 5,000,000, out of a population of 21,000,000 in Sri Lanka.
[a] The number of potential customers would be 7,000 (e.g., 15,000 total customers – 8,000 existing customers).

10. **Market Size for net accounting.** For the net accounting model, since it was introduced only in 2016, there has been still limited market data to ascertain the number of household customers to install rooftop solar PV systems under the net accounting scheme.

11. **Market Size for net plus.** The net plus model would be attractive the most to industrial and commercial customers with existing low electricity grid tariff and large rooftop spaces.[10] At

[10] Many industrial, commercial, and institutional (government, hospital, and school) properties have large rooftop spaces and enjoy lower tariff rates than SLRs22.0 per kWh (see Table 1).

the lower system unit cost of $1,000 per kWp,[11] the total market size under the net plus scheme would be up to $25 million. This is based on 50 kWp systems per each industrial and commercial property, which is a very conservative estimate. At 15% of banking lending rate with 5–7 years' tenor, the total market demand for funds would be $25 million (see Table 5).

Table 5: Potential Market Size under the Net Plus Scheme

Total Commercial Customers[a]	Market Bank Lending Rates	Potential Customers	Potential Solar PV Customers	CAPEX Unit Cost	System Size (kWp)	System Cost per Customer	Market Size ($ million)
600,000	15%	<0.1%	500	$1,000	50	$50,000	25
600,000	12%	0.2%	1,500	$1,000	50	$50,000	75
600,000	10%	0.5%	3,000	$1,000	50	$50,000	150

CAPEX = capital expenditure, kWh = kilowatt-hour, kWp = kilowatt peak, n/a = not available.

[a] According to PUCSL, the total commercial entities include industrial (50,000), hotel (500), and general purpose (550,000)

D. Conclusion

12. Across the different customer segments of varying electricity consumption (and associated tariff) levels, the analysis above points to a sufficient market demand of six times of the $50 million ADB loan amount under the net metering and net plus schemes. Even at current market conditions of about 15.0% of commercial bank lending rate with a 5–7 years of tenor, the market demand of $35 million under the net metering scheme (Table 2) could absorb 70% of available funds. The available market demand is also based on conservative estimates of only 2–20% of potentially qualifying residential customers under each tariff category (or fewer than 0.6% of the total residential customers) and only 0.1–0.5% of potentially qualifying commercial customers. There is a lack of market data to determine the exact size of the customers who would choose the net accounting scheme. It is expected that commercial and industrial customer under the net plus scheme would account for about 40% of the market demand for ADB funds, and the remaining households that consume more than 120 kWh-month (under net metering scheme) would likely account for the remaining market demand of ADB funds, depending upon the bank lending rates.

[11] In this case, the much greater economy of scale brings down the system unit costs from $1,500 for residential and small rooftop solar systems to $1,000 for large rooftop solar system on industrial, commercial, and institutional properties. In terms of the unit cost, it is between $0.87–$0.13 (SLRs13.33–SLRs20.00) per kWh. The cost estimation is conservative, as similar costs are higher in other countries.

INDICATIVE SUBPROJECT PIPELINE – CONFIDENTIAL

This information has been removed as it falls within exceptions to disclosure specified in paragraph 97, (v) of ADB’s Public Communications Policy (2011).